

AI Voice Receptionist for Brightsmile Dental

Prepared for Brightsmile Dental Group · March 14, 2026

Client	Brightsmile Dental Group (3-location practice, Hamburg DE)
Engagement	Fixed-price, milestone-based
Timeline	5 weeks from kickoff to go-live
Investment	€3,800 (build) + €180/month (managed support)
Compliance	GDPR, healthcare data minimisation, vendor DPA chain

Executive Summary

Brightsmile Dental currently loses an estimated 22% of inbound call volume to voicemail during peak hours, with the front desk handling roughly 140 calls per day across three locations. This proposal outlines a focused engagement to deploy an AI voice receptionist that answers every call within two rings, books and reschedules appointments against your existing Dampsoft DS-Win-Plus calendar, and escalates dental emergencies to on-call staff with a warm transfer.

Our approach is deliberately lean. Rather than rebuilding your telephony stack, we layer the AI agent on top of your existing Sipgate trunk and route only specific intents — booking, rescheduling, opening hours, insurance questions — to the AI. Anything outside that scope, including pain reports, medication questions, or distressed callers, is handed off to a human immediately. This keeps the clinical risk surface small and the build cost modest.

Understanding of Requirements

Your brief identified three concrete pain points: missed calls during lunch and end-of-day windows, repetitive booking calls that consume front-desk time better spent with in-person patients, and after-hours coverage where the answering service has a 14% no-show-to-callback rate. We have prioritised solutions for each.

We also note three constraints that shape the build. First, all patient data must remain inside the EU and processed only by vendors with whom Brightsmile holds a Data Processing Agreement. Second, the AI must never give clinical advice — pain triage, medication, post-op questions are out of scope and route to a human. Third, the system must speak natural German with optional English fallback, since 18% of your patient base is international.

Proposed Solution

Voice & telephony layer

Inbound calls hit your existing Sipgate number unchanged. A SIP redirect routes the audio stream to our agent service, which uses Deepgram Nova-2 for German speech-to-text (95.4% word accuracy on our healthcare

benchmark) and ElevenLabs Multilingual v2 for natural-sounding voice output. End-to-end response latency targets 1.4 seconds at the 95th percentile.

Conversation logic

We use Claude Sonnet 4 as the core reasoning engine, with a tightly scoped system prompt and a small set of tool definitions: **check_availability**, **book_appointment**, **reschedule_appointment**, **cancel_appointment**, **answer_faq**, and **escalate_to_human**. The model has no free-form clinical reasoning capability — every action is a tool call against a deterministic backend.

Calendar integration

Your Dampsoft instance exposes a CalDAV interface, which we will consume directly. Bookings are written with a 'PROVISIONAL' status until confirmed, and a nightly reconciliation job catches any conflicts created by manual front-desk edits.

Feature Breakdown

Feature	Priority	Approach
German voice (Hamburg dialect tolerant)	Must	ElevenLabs custom voice + Deepgram DE
Booking, rescheduling, cancellation	Must	Tool calls against Dampsoft CalDAV
Opening hours & location FAQ	Must	RAG over your own clinic FAQ document
Insurance question handling	Must	Scripted answers — TK, AOK, Barmer, IKK, BKK, private
Pain / emergency escalation	Must	Keyword + intent classifier → warm transfer to on-call
SMS reminders 24h and 2h before	Must	Sipgate SMS API
English fallback	Should	Same agent, language detected from caller
Voicemail transcription to email	Should	After-hours fallback path
Recall campaigns (6-month checkup)	Nice	Outbound pilot, post-launch

Relevant Experience

Nexrai Labs has delivered three voice agents in regulated settings in the past 18 months: a physiotherapy clinic in Munich (8% lift in appointment fill rate), a tax advisor in Frankfurt (60% reduction in repeat status calls), and a small veterinary chain in Cologne (after-hours triage, with hard escalation rules). The dental domain is new for us, but the safety patterns — narrow tool surface, no clinical reasoning, fast handoff — transfer directly.

Project Timeline

Week	Phase	Deliverable
1	Discovery & FAQ ingestion	Approved knowledge base, voice persona signed off
2	Calendar integration	End-to-end booking in staging against test calendar
3	Conversation tuning	20 recorded test calls, error rate under 5%

4	Pilot at one location	Live on Hamburg-Altona for one week, daily review
5	Rollout & handover	All three locations live, staff trained, runbook delivered

Methodology

We work in weekly sprints with a Friday demo call. Every recorded call from the pilot week is reviewed by both teams against a rubric covering accuracy, tone, escalation correctness, and GDPR compliance. The system is not promoted to the second or third location until the pilot location has logged 100 successful calls with zero clinical-advice incidents.

Pricing

Item	Cost
Discovery & integration build (one-time)	€2,400
Voice persona, FAQ ingestion, prompt tuning	€900
Pilot, rollout, staff training (one-time)	€500
Managed support (per month, 12-month minimum)	€180
Estimated infrastructure (Deepgram + ElevenLabs + Claude API)	€90–€140 / month

Why Nexrai Labs

- Lean, scope-disciplined builds — we do not propose enterprise architecture for SMB engagements.
- Healthcare-aware safety patterns: narrow tool surface, no clinical reasoning, hard escalation rules.
- EU-based infrastructure with vendor DPAs already in place (Deepgram EU, ElevenLabs EU, Anthropic EU endpoint).
- Three years of voice-agent experience in regulated SMB settings.
- Single point of contact through the engagement — no account-manager layer.

Appendix

References available on request: Praxis am Englischen Garten (Munich), Steuerkanzlei Bauer (Frankfurt), TierMed Köln. Architecture diagram, sample call transcripts, and the draft Auftragsverarbeitungsvertrag (AVV) are attached to this proposal in the shared folder.